

Multiform Optimization for Cross-Modality Attacks in Person Re-Identification

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Abstract

In recent years, despite significant advancements in adversarial attack research, the security challenges in cross-modal scenarios, such as the transferability of adversarial attacks between infrared, thermal, and RGB images, have been overlooked. These heterogeneous image modalities collected by different hardware devices are widely prevalent in practical applications, and the substantial differences between modalities pose significant challenges to attack transferability. In this work, we explore a novel cross-modal adversarial attack strategy, termed multiform attack. We propose a dual-layer optimization framework based on gradient-evolution, facilitating efficient perturbation transfer between modalities. In the first phase of optimization, the framework uses image gradients to learn universal perturbations within each modality. Then, through a second optimization phase, it introduces evolutionary algorithms to search for cross-modally transferable shared perturbations, thereby enhancing the transferability of the universal perturbations. Through extensive testing on multiple heterogeneous datasets, we demonstrate the superiority and robustness of Multiform Attack compared to existing techniques. This work not only enhances the transferability of cross-modal adversarial attacks but also provides a new perspective for understanding security vulnerabilities in cross-modal systems. The code will be available.

1. Introduction

Person re-identification (ReID) [31] is a critical technology that identifies individuals across cameras and time intervals by analyzing surveillance images. Ensuring the security of ReID systems is of paramount importance. For instance, attackers could embed adversarial perturbations into clothing or stickers, disrupting the images captured by cameras and significantly reducing the recognition accuracy of intelligent systems [10].

Most existing studies focus on the security of visible-light ReID systems. However, in real-world applications, ReID systems often employ cameras with different modal-

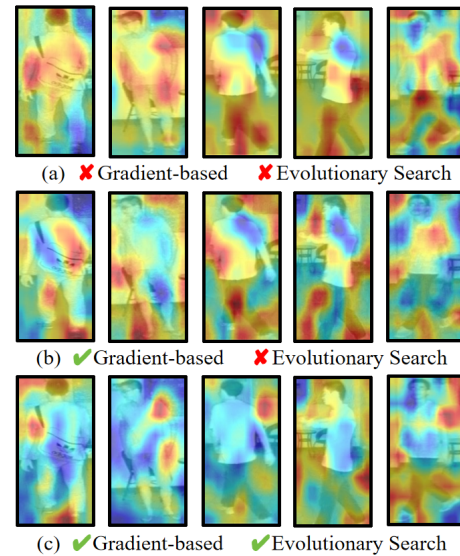


Figure 1. Comparison of attention heatmaps for transfer attacks (from LLCM [28] to SYSU [24] examples) demonstrates how attention becomes increasingly dispersed through the joint optimization of two types in the multiform attack, highlighting its effectiveness in enhancing transfer attacks.

ities (e.g., RGB, infrared, and thermal) to adapt to diverse environmental conditions. Furthermore, with increasingly stringent privacy regulations, personal data processing technologies face higher compliance requirements. To address this, some ReID systems utilize special image transformation methods or leverage multimodal data to enhance privacy protection [11, 19].

Does the increased complexity of cross-modal ReID systems result in enhanced security? The answer is no. Despite their sophisticated design, cross-modal ReID systems do not demonstrate significantly improved security. Recent research [7] highlights this issue. However, current research have largely overlooked the transferability of adversarial perturbations in cross-modal scenarios, leaving critical mechanisms and impacts insufficiently explored.

A systematic investigation into the security vulnerabilities of cross-modal ReID systems is essential for identifying

their weaknesses and enhancing robustness (refer to the adversarial defense analysis in the supplementary materials). Such research not only advances the field of adversarial attacks but also provides a theoretical foundation and practical insights for developing more secure multimodal intelligent systems.

Currently, ReID attack methods can be categorized into two types: single-modality attacks [6, 30] and cross-modality attacks [7]. Single-modality attacks focus on images within the same modality, such as RGB-to-RGB. While effective within a single domain, these methods lack adaptability to cross-modal scenarios. In contrast, cross-modality attacks aim to transfer perturbations across different modalities. The core challenge here is the substantial feature differences between modalities, which complicate ensuring both effectiveness and transferability. In this paper, we focus on cross-modality attacks. Our goal is to find a universal perturbation that captures shared features between modalities, enabling more effective and broader transferability across diverse scenarios.

However, achieving this objective presents challenges. Like model overfitting, adversarial perturbations can become overly specialized to modality-specific features, resulting in poor performance on unseen modalities. One potential solution is to train models independently for each modality and use their gradients to learn a universal perturbation. However, inconsistencies in gradients, resulting from differences in model architectures and heterogeneous training data, make it difficult to effectively form a unified perturbation. Moreover, variations in model performance across modalities may lead to imbalanced perturbation learning, adversely affecting generalization and universality.

These challenges introduce inherent conflicts in the optimization process—where improving transferability for one modality may degrade performance for another. Therefore, generating universal and transferable perturbations for cross-modal adversarial attacks in person re-identification inherently becomes a multi-objective optimization problem. The core challenge lies in balancing transferability and perturbation magnitude to ensure robust performance across all modalities.

To address these challenges, we propose using evolutionary search due to its natural advantage in handling complex optimization tasks. As a global optimization method, evolutionary search explores a broad search space and effectively manages conflicting objectives. This makes it particularly suitable for our problem, as it can balance maximizing transferability and minimizing perturbation magnitude, thereby capturing shared perturbation features across modalities. The generated universal perturbations ultimately maintain consistent, high performance across all modalities, enabling effective attacks on unseen ones.

However, relying solely on evolutionary search to optimize individual images can be computationally expensive, especially in high-dimensional spaces like images. To address this issue, we combined gradient-based optimization, first identifying a good solution in a specific modality and then using it as the initialization for evolutionary search. This approach effectively narrows the search space and reduces computational complexity. By combining gradient optimization with evolutionary search, we achieve high transferability while optimizing computational performance, resulting in an efficient and effective balance. As shown in Fig.1, the gradient-based optimization method demonstrates moderate transferability compared to the attention heatmaps of clean samples (Fig.(a) vs. Fig.(b)). With the integration of diverse optimization mechanisms, the attack’s transferability is significantly enhanced (Fig.(c)).

Specifically, we propose a Gradient-Evolutionary Multimodal Optimization framework to ensure effective and transferable adversarial attacks across heterogeneous modalities. We leverage evolutionary computation to search for sparse perturbations applicable to different modalities and use these perturbations to fine-tune a universal perturbation, thereby enhancing its transferability. The first optimization phase uses a gradient-based method to optimize the perturbation within a specific modality, maximizing its impact. The second phase applies evolutionary search to find perturbations that efficiently transfer across different modalities, aiming to achieve broadly applicable solutions.

Our main contributions can be summarized as:

- We propose a dual-layer optimization framework combining gradient-based and evolutionary search techniques for cross-modal adversarial attacks. This multimodal optimization enables effective knowledge transfer across tasks, significantly improving attack effectiveness and transferability.
- We incorporate evolutionary algorithms into adversarial attacks and, through mathematical analysis, demonstrate their effectiveness in enhancing cross-modal transferability and addressing complex constraints, providing a solid theoretical foundation for multi-modal scenarios.
- Extensive experiments validate the superiority of our method, showing clear improvements over existing approaches in transferability and robustness, advancing the understanding of security in multi-modal systems.

2. Related Work

2.1. Adversarial Attack

The concept of adversarial attacks was first introduced by Szegedy et al. [8], whose research revealed that even small perturbations to input images could mislead deep neural networks, resulting in incorrect image recognition. This

finding not only highlights the vulnerability of deep learning models but also has important theoretical and practical implications for enhancing the security of artificial intelligence systems. Subsequently, a plethora of adversarial attacks have been proposed [3, 12, 16]. The work by Moosavi-Dezfooli et al. introduced universal adversarial perturbations [17], further advancing research in this field. They demonstrated the ability to generate nearly 'universal' perturbation vectors that, when added to any data sample, cause the same deep learning model to produce incorrect outputs. One Pixel Attack [20], as a significant milestone in sparse perturbation attacks, demonstrates the possibility of misleading models by modifying a single pixel in an image. However, the modification of individual pixels may not always successfully attack all types of images or models in real applications. Therefore, sparse adversarial attacks [4, 20, 22, 23] often involve modifications of multiple pixels, albeit still limited in number, providing higher flexibility and a wider success rate. Although universal perturbations can broadly affect multiple samples, they are relatively easier to be detected by designed targeted detection mechanisms due to their ubiquitous and consistent perturbation patterns. In contrast, sparse adversarial attacks, by applying extremely limited perturbations to input data, demonstrate higher stealthiness. This attack method is more challenging to be identified by standard defense measures in experiments and practical applications due to its high target accuracy and fewer intervention points.

2.2. Attack Person Re-ID System

Several ReID attack methods have been proposed, with current research predominantly focusing on RGB-RGB matching. These methods include: Metric-FGSM [1] extends techniques inspired by classification attacks into the category of metric attacks. These include Fast Gradient Sign Method (FGSM)[8], Iterative FGSM (IFGSM), and Momentum IFGSM (MIFGSM)[5]. The Furthest-Negative Attack (FNA)[2] integrates hard sample mining[9] and triplet loss to guide image features towards the least similar cluster while moving away from similar features. Deep Mis-Ranking (DMR) [21] utilizes a multi-stage network architecture to extract features at different levels, aiming to derive general and transferable features for adversarial perturbations. Gong et al.[6] proposed a method specifically for attacking color features without requiring additional reference images and discussed effective defense strategies against current ReID attacks. The Opposite-Direction Feature Attack (ODFA)[30] exploits feature-level adversarial gradients to generate examples that guide features in the opposite direction using an artificial guide. Yang et al.[25] introduced a combined attack named Col.+Del. (Color Attack and Delta Attack), which integrates UAP-Retrieval [14] with color space perturbations [13]. The in-

clusion of color space perturbations enhances the attack's universality and transferability across RGB-RGB datasets. CMPS [7] represents the first exploration into the security of cross-modal ReID. It leverages gradients from different modalities to optimize universal perturbations, effectively enhancing the universality and adaptability of attacks within a given modality. Similar to other gradient-based methods, it has certain limitations in terms of the transferability of attacks.

Existing methods primarily focus on gradient-based attacks for single-modal systems, lacking mechanisms to capture shared knowledge across different modalities. Additionally, the heterogeneity between different modalities makes it difficult for these gradient-based methods to achieve effective adaptation across more than two modalities. Our approach aims to enhance the effectiveness and transferability of cross-modal adversarial attacks by combining gradient-based techniques with evolutionary search.

3. Methodology

We aim to find a universal adversarial perturbation ϵ with cross-modal transferability that can mislead the ranking results of a given modality $\mathcal{G}$ and an unseen target modality χ for a re-identification (re-ID) model. The attack involves modifying a query image $\mathcal{I}$ by adding a perturbation ϵ . This perturbed image $\mathcal{I}'$ is then used to fool the victim re-identification model $\mathcal{M}$ when querying a gallery.

3.1. Framework Overview

Our proposed methodology employs a dual-layer optimization framework, integrating gradient-based learning and evolutionary algorithms to enhance the effectiveness and transferability of adversarial perturbations across different image modalities. This framework is designed to address the unique challenges posed by the heterogeneity of cross-modal data, ensuring that the learned perturbations are both effective and broadly applicable. In the first layer of optimization, a gradient-based learning method focuses on optimizing adversarial perturbations to attack machine learning models within specific modalities. This process involves computing the loss based on the task-specific metric, such as the triplet loss with Mahalanobis distance, and using momentum gradient descent to iteratively adjust the perturbations. The second layer of optimization employs an evolutionary search strategy to explore perturbations that can be effectively transferred between models trained on different modalities. This strategy involves generating a population of perturbations, evaluating their performance across multiple models, and iteratively refining the perturbations through crossover and mutation operations. The goal is to discover perturbations that are broadly applicable and maintain their effectiveness across various modalities. By leveraging evolutionary computation, this layer addresses the

challenge of transferring adversarial attacks between heterogeneous data, enhancing the robustness and generalization of the perturbations.

The combination of these two layers—gradient-based learning for modality-specific optimization and evolutionary search for cross-modal transferability—constitutes the Gradient-Evolutionary Multimodal Optimization framework. This dual-layer approach not only optimizes perturbations for a single modality but also adapts them to the security challenges present in multi-modal environments. The overall framework is detailed in Alg.1 in the supplementary materials. This algorithm delineates the step-by-step process for implementing the Gradient-Evolutionary Multimodal Optimization, ensuring continuous refinement and adaptation of perturbations to maintain high effectiveness across different image modalities. Regarding the proposed method, we conducted a theoretical analysis focusing on two aspects: the feasibility of evolutionary search and its effectiveness in enhancing the transferability of universal perturbations. For details, please refer to Supplementary Materials.

3.2. Gradient-Based Learning

In the first layer, our primary objective is to optimize adversarial perturbations for specific modalities using gradient-based learning. We define the optimization problem as:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\text{tri}}(\delta, x_i), \quad (1)$$

where $\mathcal{L}_{\text{tri}}$ is the loss function tailored to optimize adversarial perturbations within a specific modality.

Existing research [1, 2, 6, 7, 21, 25, 30] on ReID adversarial attacks typically employs Euclidean distance to design loss functions. However, under the assumption that adversarial samples reside in a manifold space, traditional Euclidean distance may not be sufficiently flexible, as data points in manifold spaces often exhibit nonlinear distributions. To address this issue, we employ the Mahalanobis distance, which is more suitable for manifold problems. This distance measure considers the covariance structure of the data, allowing it to more accurately capture the nonlinear relationships and adapt to scale variations in different directions, thereby providing a more flexible and precise distance metric. Hence, in crafting the triplet loss function, we opt to utilize Mahalanobis distance as our metric of choice, aiming to better guide the optimization process for adversarial perturbations:

$$\begin{aligned} \mathcal{L}_{\text{tri}}(\delta, x_i) = & [D_M(C_n^{m_1}, f_{\text{adv}}) - D_M(C_p^{m_1}, f_{\text{adv}}) + \rho]_+ \\ & + [D_M(C_n^{m_2}, f_{\text{adv}}) - D_M(C_p^{m_2}, f_{\text{adv}}) + \rho]_+. \end{aligned} \quad (2)$$

We optimize the perturbation using cluster centroids [14]. Here, $C_p^{m_1}$, $C_p^{m_2}$, $C_n^{m_1}$, and $C_n^{m_2}$ respectively

represent the nearest and farthest cluster centroids of original image features in the training data for modalities m_1 and m_2 . f_{adv} denote the perturbed features (We set the margin $\rho = 0.5$ in triplet loss). The distance between vectors x and y , using the Mahalanobis distance $D_M(x, y)$, is defined as follows. For computational convenience and optimization stability, the squared Mahalanobis distance is commonly employed as the loss function:

$$D_M(x, y) = (x - y)^T S^{-1} (x - y), \quad (3)$$

Here, S is the covariance matrix of the dataset. We utilize exponential weighted moving average for momentum gradient descent. This approach facilitates smoother parameter updates, accelerating convergence and enhancing generalization performance. The process is formulated as follows:

$$v_{t+1} = \beta v_t + (1 - \beta) \cdot \frac{\nabla_{\delta} \mathcal{L}}{\|\nabla_{\delta} \mathcal{L}\|_1}. \quad (4)$$

Here, v_t represents the exponential moving average of the gradient at time step t (initial value $v_0 = 0.$), β is the decay coefficient (set as $\beta = 0.9$), and $\frac{\nabla_{\delta} \mathcal{L}}{\|\nabla_{\delta} \mathcal{L}\|_1}$ is the normalized gradient. Then, use the updated momentum variable v_{t+1} to update the perturbation δ :

$$\delta_{t+1} = \text{clip}(\delta_t + \alpha \cdot \text{sign}(v_{t+1}), -\epsilon, \epsilon). \quad (5)$$

Here, δ_{t+1} is the updated perturbation at time step $t + 1$, δ_t is the perturbation at time step t , α is the learning rate (set as $\alpha = \frac{\epsilon}{10}$), ϵ is the clipping threshold ($\epsilon = 8$, unless otherwise specified), and $\text{sign}(\cdot)$ function returns the sign of the input.

In this layer of optimization, we focus on minimizing the task-specific loss, which aims to mislead the ReID model by altering the query image such that it fails to match the correct individual in the database.

3.3. Evolutionary Search

In the second optimization layer, we employ evolutionary search to enhance the transferability of perturbations across different modalities. This approach leverages evolutionary algorithms to simultaneously search for sparse perturbations with cross-modal transferability, which are then used to fine-tune the universal adversarial perturbations obtained from the first optimization layer. With respect to our objectives, we define the optimization problem as follows:

$$\begin{aligned} \min_{\eta} \quad & \Phi(\mathcal{F}(x + \delta + \eta)), \\ \text{subject to:} \quad & \|\eta\|_0 \leq k, \quad \|\delta + \eta\|_{\infty} \leq \epsilon. \end{aligned} \quad (6)$$

k represents the number of perturbed pixels. The $\|\delta + \eta\|_{\infty}$ constrains that the maximum value of each element in the perturbation vector does not exceed ϵ , which can be achieved through clipping.

At this stage, the optimized adversarial sample x_{adv} can be obtained from $x + \delta + \eta$. The features of the adversarial sample, denoted as f_{adv} , are extracted using $\mathcal{F}(x_{adv})$. Where $\Phi(f_{adv}) = (\tilde{\mathcal{D}}(f_{adv}), \tilde{\mathcal{S}}(f_{adv}), \|\eta\|_2, \|\eta\|_0)^T$ is the objective vector. To address the limitations of using $\{-1, 1\}$ as the perturbation set, we redefine it as $\{-1, 0, 1\}$. This expansion allows the inclusion of zero values within the perturbation vector, inherently optimizing the l_0 norm by increasing the proportion of zeros. Consequently, optimizing the l_2 norm also indirectly reduces the l_0 norm, as a lower l_2 norm can be achieved partly by increasing the number of zeros in the vector. Therefore, the objective vector $\Phi(f_{adv})$ is now defined as $(\tilde{\mathcal{D}}(f_{adv}), \tilde{\mathcal{S}}(f_{adv}), \|\eta\|_2)^T$, where $\eta \in \{-1, 1, 0\}$. $\tilde{\mathcal{D}}(f_{adv})$ and $\tilde{\mathcal{S}}(f_{adv})$ represent the total metric loss and total attack success rate across all models, respectively. They can be described by the following formula:

$$\mathcal{D}_i(f_{adv}) = (f_{adv} - C)^T S^{-1} (f_{adv} - C), \quad (7)$$

$\mathcal{D}_i(f_{adv})$ denote the loss incurred by the perturbed input on model $\mathcal{M}_i$. The loss from the adversarial sample x_{adv} to the cluster centroid C is measured using the squared Mahalanobis distance. Therefore, the total loss across all models can be represented as:

$$\mathcal{D}(f_{adv}) = \sum_{i=1}^n \mathcal{D}_i(f_{adv}). \quad (8)$$

To transform into a minimization problem, we ultimately use the following formula for optimization:

$$\tilde{\mathcal{D}}(f_{adv}) = \exp(-\mathcal{D}(f_{adv})). \quad (9)$$

For model $\mathcal{M}_i$, success rates $\mathcal{S}_i$ can be defined as follows:

$$\mathcal{S}_i(f_{adv}) = \begin{cases} 1, & \text{if } \arg \max(\hat{y}_j) \neq y_j \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

$\hat{y}_j$ is the predicted label by the model, and y_j is the true label corresponding to the sample. The overall success rate can be calculated using the following formula:

$$\mathcal{S}(f_{adv}) = \frac{1}{n} \sum_{i=1}^n \mathcal{S}_i(f_{adv}). \quad (11)$$

To transform into a minimization problem, we ultimately use the following formula for optimization:

$$\tilde{\mathcal{S}}(f_{adv}) = 1 - \frac{1}{n} \sum_{i=1}^n \mathcal{S}_i(f_{adv}). \quad (12)$$

We follow the approach proposed by Williams et al. [23] for crossover, mutation, and evaluation of the population.

For further details, please refer to the supplementary materials. During the selection phase, we define the following non-dominated sorting relationship to achieve the objective of simultaneously searching for transferable perturbations across multiple modalities.

Domination Definition. In the process of conducting multimodal adversarial attacks, we assess and compare two perturbation sets within the perturbation solution set $\mathcal{P}$, denoted as $\mathcal{P}_i$ and $\mathcal{P}_j$, respectively. These two solutions yield perturbations represented by η_i and η_j . We evaluate the resulting adversarial effectiveness using function $\mathcal{F}(\bullet)$ which yields the objective vectors $\mathcal{F}_i$ and $\mathcal{F}_j$. A solution $\mathcal{P}_i$ is considered to dominate another solution $\mathcal{P}_j$ if any of the following conditions are met:

1. If η_i has higher transferability than η_j .
2. η_i and η_j have the same transferability, and $\|\eta_i\|_2 \leq \|\eta_j\|_2$.
3. Both η_i and η_j do not exhibit transferability, and η_i has a smaller total loss $\tilde{\mathcal{D}}(f_{adv})$.

Please note, a perturbation η is considered to have transferability if it satisfies the attack success rate is greater than 0. The higher the attack success rate $\mathcal{S}(f_{adv})$, the greater the considered transferability.

This dual-layer optimization framework significantly enhances the robustness of adversarial perturbations. The first layer of optimization utilizes gradient descent to learn universal perturbations. The second layer employs evolutionary strategies to capture transferable features across modalities, fine-tuning the learned universal perturbations. This approach not only improves the applicability of perturbations in multi-modal environments but also increases the flexibility of the overall attack strategy.

3.4. Theoretical Analysis of Attack Transferability

In this section, we discuss how evolutionary search can be leveraged to optimize attack perturbations, enhancing their transferability across different models. We define a fitness function to evaluate the effectiveness of the fine-tuned perturbation δ_f as follows:

$$f(\delta_f) = \sum_{i=1}^k w_i \cdot r_i(\delta_u + \delta_f) - \lambda \cdot \|\delta_f\|_0. \quad (13)$$

Here, $r_i(\delta_u + \delta_f)$ is the misclassification rate for model $\mathcal{M}_i$ when both the universal perturbation δ_u and fine-tuned perturbation δ_f are applied. The term $\|\delta_f\|_0$ measures the sparsity of the perturbation, with w_i and λ being the model weight and sparsity regularization parameter, respectively. The objective of this function is to maximize the effectiveness of δ_f across multiple models while maintaining its sparsity.

To measure how well δ_f complements δ_u , we introduce the complementarity measure α_i :

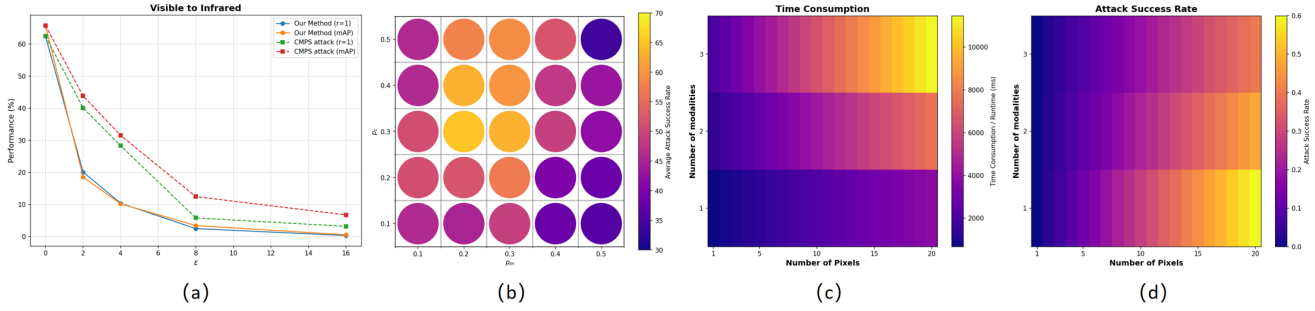


Figure 2. (a) Performance comparison under different perturbation sizes (ϵ) between our method and the state-of-the-art attack CMPS. (b) The impact of mutation rates (p_m) and crossover rates (p_c) on the attack success rate during evolutionary search. (c) The relationship among the number of perturbed pixels, the number of modalities, and time consumption. (d) The interplay among the number of perturbed pixels, the number of modalities, and the attack success rate.

Table 1. Results for attacking cross-modality ReID systems on the RegDB dataset. It reports on visible images querying thermal images and vice versa. Rank at r accuracy (%) and mAP (%) are reported. The perturbation ϵ is set to 8.

Settings		Visible to Thermal				Thermal to Visible			
Method	Venue	$r = 1$	$r = 10$	$r = 20$	mAP	$r = 1$	$r = 10$	$r = 20$	mAP
AGW baseline [27]	TPAMI 2022	70.05	86.21	91.55	66.37	70.49	87.21	91.84	65.90
M-FGSM attack [1]	TPAMI 2020	29.34	52.90	61.44	23.35	23.64	40.36	48.61	18.57
LTA attack [6]	CVPR 2022	12.65	25.24	34.02	12.80	10.51	22.93	31.79	9.74
ODFA attack [30]	IJCV 2023	28.57	51.42	60.58	21.84	17.26	33.27	42.92	15.27
Col.+Del. attack [25]	TPAMI 2023	5.12	16.83	22.10	4.94	4.92	14.47	23.04	4.86
CMPS attack [7]	NeurIPS 2024	2.29	9.06	18.35	3.92	1.93	11.44	19.30	3.46
Our attack		0.98	6.24	9.27	1.26	1.04	7.47	10.02	1.31

4. Empirical Study

4.1. Dataset

We evaluate our method on four cross-modality ReID datasets: RegDB [18], LLCM [28], SYSU [24], and CnMix. SYSU includes 395 training identities captured by six cameras (four RGB and two near-infrared), with 22,258 visible and 11,909 near-infrared images, and a test set of 95 identities with 3,803 query images. LLCM is designed for low-light cross-modality ReID, presenting diverse scenarios and challenges for adversarial attacks. CnMix, created by applying random channel mixing to Market1501 [29] images, simulates privacy-preserving image transformations (details in supplementary materials).

4.2. Evaluation Metric

Following [29], we use Rank-k precision, Cumulative Matching Characteristics (CMC), and mean Average Precision (mAP) as evaluation metrics. Rank-1 measures the accuracy of the top result, while mAP evaluates the mean accuracy based on ranked query results. In adversarial attacks, lower accuracy scores indicate higher attack effectiveness.

$$\alpha_i = \frac{r_i(\delta_u + \delta_f) - r_i(\delta_u)}{r_i(\delta_u)}, \quad (14)$$

this measure quantifies the increase in misclassification rate when δ_f is added to δ_u . Ideally, α_i should be significantly positive, indicating that δ_f effectively enhances δ_u .

Through evolutionary search involving selection, crossover, and mutation, we can iteratively optimize δ_f . It can be proven that the complementarity measure α_i increases with each iteration:

$$\alpha_i^{(t+1)} \geq \alpha_i^{(t)}, \quad (15)$$

this indicates that the effectiveness of the perturbation improves with each iteration. Furthermore, the non-decreasing nature of the fitness function ensures the convergence of the evolutionary search process.

In summary, this analysis demonstrates that evolutionary search can effectively optimize δ_f , enhancing the transferability of attacks across multiple models and providing critical theoretical support for improving security in multimodal systems. Detailed derivation can be found in the supplementary materials.

Table 2. Results for attacking cross-modality ReID systems on the LLCM [28] dataset. It reports on visible images querying thermal images and vice versa. Rank at r accuracy (%) and mAP (%) are reported.

Settings		Visible to Infrared				Infrared to Visible			
Method	Venue	$r = 1$	$r = 10$	$r = 20$	mAP	$r = 1$	$r = 10$	$r = 20$	mAP
DEEN baseline [27]	CVPR 2023	62.53	90.31	94.73	65.84	54.96	84.92	90.91	62.95
FGSM attack [8]	ICLR 2015	54.64	78.23	88.76	50.32	49.18	73.42	86.12	47.89
PGD attack [15]	ICLR 2018	49.14	70.34	81.54	45.76	44.25	68.39	80.12	42.81
M-FGSM attack [1]	TPAMI 2020	28.48	64.92	75.12	32.88	25.64	61.45	78.31	30.46
LTA attack [6]	CVPR 2022	15.16	56.42	67.53	21.47	19.54	58.25	70.72	24.86
ODFA attack [30]	IJCV 2023	26.34	65.24	76.92	30.85	23.73	62.46	73.57	29.63
CMPS attack [7]	NeurIPS 2024	5.83	18.14	27.56	12.47	6.42	19.53	28.54	12.23
Our attack		2.49	10.56	15.77	3.43	3.61	11.27	17.33	4.92

Table 3. Performance comparison with the state-of-the-art method in Visible to Non-visible and Non-visible to Visible scenarios.

Cross-Dataset Testing	Attack Type	$r=1$ (%)	mAP (%)	Method	Attack Type	$r=1$ (%)	mAP (%)
SYSU $\rightarrow$ CnMix	CMPS [7]	13.83 ± 0.35	12.29 ± 0.28	CnMix $\rightarrow$ SYSU	CMPS [7]	15.75 ± 0.38	14.41 ± 0.30
SYSU $\rightarrow$ CnMix	Ours	6.81 ± 0.20	6.02 ± 0.15	CnMix $\rightarrow$ SYSU	Ours	8.23 ± 0.21	7.74 ± 0.18
SYSU $\rightarrow$ Sketch	CMPS [7]	16.46 ± 0.40	14.78 ± 0.32	Sketch $\rightarrow$ SYSU	CMPS [7]	17.34 ± 0.45	16.87 ± 0.35
SYSU $\rightarrow$ Sketch	Ours	7.14 ± 0.22	6.75 ± 0.19	Sketch $\rightarrow$ SYSU	Ours	9.38 ± 0.24	8.63 ± 0.20

(a) Visible to Non-visible (b) Non-visible to Visible

4.3. Comparison

Following [7], we employed cross-modality baseline models, AGW [27], DEEN [28] and DDAG [26], to evaluate the performance of the attack methods. The experiments comprised two scenarios: 1) Perturbing visible images (query) to disrupt the retrieval of infrared or thermal images (gallery). 2) Perturbing infrared or thermal images (query) to interfere with the retrieval of visible images (gallery).

Tab.1 and Tab.2 compare cross-modality attack performance on the LLCM and RegDB datasets. Results show that our method consistently achieves the lowest Rank-1 accuracy and mAP across all settings, significantly outperforming other methods. In contrast, traditional approaches like FGSM [8], PGD [15], LTA [6], and ODFA [30], while effective in single-modality tasks, are limited in cross-modality scenarios due to their inability to fully exploit shared features between modalities. CMPS [7], as a cross-modality attack method, enhances performance through gradient-based optimization but struggles with significant modality distribution differences, which hinder its convergence to a global optimum. These results underscore the clear advantages of our approach.

In Fig. 2(a), we present the performance comparison between our proposed method and the state-of-the-art CMPS [7] under different perturbation sizes ϵ , using the DEEN model evaluated on the LLCM dataset. The experimental results demonstrate that as ϵ decreases, the performance gap between our method and CMPS widens significantly. In particular, under small perturbations, our method

outperforms CMPS in both Rank-1 accuracy and mAP. This highlights the advantages of the evolutionary search strategy, which effectively optimizes shared perturbations across modalities under multi-objective constraints. Consequently, our method achieves higher attack efficiency and transferability under low-perturbation conditions. These results indicate that our method not only ensures effective attacks but also exhibits superior robustness and adaptability.

Experiments Comparing Attack Transferability. We verify transferability using three heterogeneous datasets: SYSU, RegDB, and CnMix. In our transfer attack experiments, all datasets are used simultaneously, with each dataset trained on a different model representing a specific modality. For instance, in Tab.3, SYSU $\rightarrow$ CnMix indicates that we optimize the universal perturbation on the SYSU dataset (using RegDB for auxiliary optimization) and then transfer it to CnMix for testing. This part of the experiment involves three modalities: RGB, infrared, and thermal. For a fair comparison, CMPS is optimized simultaneously across all three modalities in the involved datasets.

It can be observed that CMPS, which relies on gradient optimization across multiple modalities, may suffer from inconsistent gradient directions due to significant feature distribution differences between modalities, leading to sub-optimal results. In contrast, the proposed method demonstrates clear superiority. This indicates: 1) Perturbations trained solely on gradients face challenges due to inconsistent gradient information from diverse model architectures and heterogeneous training data, making effective gradient utilization difficult. Additionally, performance differences between models trained on different modalities may

Table 4. Comparison of transfer attack performance between our method and state-of-the-art methods across models in terms of Rank-1 (%) on the LLCM dataset.

Source \ Target Model	DDAG (CMPS/Ours)	AGW (CMPS/Ours)	DEEN (CMPS/Ours)
DDAG [26]	4.72 ± 0.22 / 1.28 ± 0.11	6.72 ± 0.30 / 1.81 ± 0.12	9.66 ± 0.40 / 1.25 ± 0.08
AGW [27]	5.43 ± 0.24 / 1.33 ± 0.09	3.46 ± 0.18 / 1.97 ± 0.11	8.89 ± 0.35 / 1.68 ± 0.10
DEEN [28]	4.72 ± 0.20 / 1.92 ± 0.15	5.58 ± 0.26 / 1.75 ± 0.13	5.83 ± 0.28 / 2.49 ± 0.16

Table 5. Ablation studies on the DEEN baseline using the LLCM dataset. Maha. refers to Mahalanobis distance, Grad. represents Gradient-based optimization, and Evol. indicates Evolutionary Search.

No.	Maha.	Grad.	Evol.	mAP	rank-1
0	×	×	×	62.53	65.84
1	×	✓	×	8.54	12.35
2	✓	✓	×	6.88	9.47
3	✓	✓	✓	2.49	3.43

lead to imbalanced perturbation learning, affecting universality and generalization. 2) Adversarial attacks face a similar dilemma between stability and adaptability in real-world scenarios. In deep learning, addressing complex tasks often involves a balance between stability and plasticity. Excessive flexibility to new data (high plasticity) can lead to ‘catastrophic forgetting,’ while insufficient adaptability (high stability) may hinder learning efficiency and the model’s generalization. The method proposed in this paper offers a potential solution to this dilemma.

Furthermore, as shown in Tab. 4, experiments across different baselines also demonstrate the superiority of our proposed method.

4.4. Ablation Study

Fig.2 (b), (c), and (d) illustrate the impact of various factors on attack performance during evolutionary search.

Fig.2(b) shows the impact of different crossover rates (p_c) and mutation rates (p_m) on the attack success rate when using evolutionary search alone. The heatmap reveals that higher mutation rates (p_m) generally improve success rates, especially when combined with moderate crossover rates (p_c). Notably, the combination of $p_m = 0.2$ and $p_c = 0.3$ achieves the highest success rate, highlighting the importance of balancing exploration (mutation) and exploitation (crossover) for optimal performance. Unless otherwise specified, we adopt this parameter combination in our experiments.

Fig.2(c) demonstrates the relationship between the number of perturbed pixels, the number of modalities, and time consumption. Key observations include: (1) The attack success rate improves as the number of perturbed pixels

increases; (2) Time consumption increases proportionally with the number of perturbed pixels; (3) Time consumption further rises with the number of modalities due to additional computational overhead; (4) The attack success rate declines as the number of modalities increases, highlighting the challenges of optimizing across diverse modalities.

Fig.2(d) illustrates the interplay between the number of perturbed pixels, the number of modalities, time consumption, and attack success rate. While increasing the number of perturbed pixels improves the attack success rate, it also significantly increases computational cost, particularly in multi-modality scenarios.

Table 5 evaluates the contribution of each component in our method. When Mahalanobis distance is not applied, Euclidean distance is used by default. No.0 represents the baseline performance without any attack. Introducing gradient-based optimization alone (No.1) slightly reduces the model’s performance, demonstrating its effectiveness in generating universal perturbations, although it is limited in capturing cross-modal features. Replacing Euclidean distance with Mahalanobis distance (No.2) further reduces the model’s performance, highlighting its advantage in handling modality-specific relationships. Finally, combining gradient-based optimization, Mahalanobis distance, and evolutionary search (No.3) achieves the most significant attack effect, emphasizing the critical role of evolutionary search in enhancing cross-modal perturbation transferability.

5. Conclusion

This paper presents a novel cross-modal adversarial attack strategy, Multiform Attack, leveraging a gradient-evolutionary multiform optimization framework to improve transferability across heterogeneous image modalities. By combining gradient-based learning with evolutionary search, our approach enhances the robustness and transferability of adversarial perturbations while effectively addressing cross-modal constraints. Experiments show our method outperforms existing techniques, advancing the performance of universal perturbations and providing valuable insights into strengthening multi-modal system security.

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Multiform Optimization for Cross-Modality Attacks in Person Re-Identification

Supplementary Material

6. Appendix

Algorithm 1 Multiform Attack Optimization Framework

```

1: Input:  $\mathcal{D}_{\text{train}}$ : Training dataset for gradient optimization,  $\mathcal{D}_{\text{evo}}$ : Training dataset for evolutionary search,  $\mathcal{M}_{\text{grad}}$ : Model for gradient optimization,  $\mathcal{M}_{\text{evo}}$ : List of models for evolutionary search,  $\epsilon$ : Perturbation limit,  $\alpha$ : Learning rate,  $\beta$ : Momentum decay coefficient, max_iter: Maximum number of iterations,  $k$ : Maximum number of perturbed pixels per iteration,  $\mathcal{P}$ : Initial population size,  $G$ : Number of generations in evolutionary search
2: Output:  $\delta$ : Optimized perturbation after all iterations
3: Initialization:  $\delta \leftarrow \text{Rand}(0, 1)$ ,  $v \leftarrow 0$   $\triangleright$  Exponential moving average of the gradient
4: for  $t = 0$  to max_iter  $- 1$  do
5:   Fetch batch  $x$  from  $\mathcal{D}_{\text{train}}$ 
6:   Compute the loss  $\mathcal{L}_{\text{meta}}(\delta, x)$  using  $\mathcal{M}_{\text{grad}}$ 
7:    $v_{t+1} \leftarrow \beta v_t + (1 - \beta) \frac{\nabla_{\delta} \mathcal{L}_{\text{meta}}}{\|\nabla_{\delta} \mathcal{L}_{\text{meta}}\|_1}$ 
8:    $\delta_{t+1} \leftarrow \text{clip}(\delta_t + \alpha \cdot \text{sign}(v_{t+1}), -\epsilon, \epsilon)$ 
9:   if  $t \bmod \text{iter} == 0$  then  $\triangleright$  Execute evolutionary search every  $\text{iter}$  iterations
10:     Fetch batch  $x_{\text{evo}}$  from  $\mathcal{D}_{\text{evo}}$ 
11:     Initialize population  $\mathcal{P}$ 
12:     Generate new individuals through crossover and mutation
13:     for each individual  $\eta \in \mathcal{P}$  do
14:       Evaluate  $\eta$  using all models in  $\mathcal{M}_{\text{evo}}$  on  $x_{\text{evo}}$ 
15:     end for
16:     Perform non-dominated sorting to select the best  $\eta$ 
17:      $\delta_t \leftarrow \delta_t + \eta$   $\triangleright$  Update  $\delta_t$  with the best  $\eta$  found
18:   end if
19: end for
20: Output the optimized perturbation  $\delta$ 

```

7. Description of Experimental Equipment

Our device is equipped with three NVIDIA A40 GPUs, each with 48GB of memory.

8. Details of the Evolutionary Algorithm

8.1. Initialization

The attack method initializes by setting the number of perturbed pixels to k and constructing a set $\mathcal{P}$ of s solutions. Each solution is created by uniformly sampling pixel loca-

tions from the set $\{1, \dots, h \cdot w\}$. The initial perturbation values for each color channel are randomly sampled from the set $\{-1, 0, 1\}$:

$$\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_s\}, \quad \mathcal{P}_i \in \{-1, 0, 1\} \quad (16)$$

where:

- $\mathcal{P}$: The set of initial solutions.
- s : The total number of solutions.

When invoking the evolutionary process, we set a fixed number of iterations to 200.

8.2. Crossover

The crossover operation aims to generate new solutions by combining traits from two parent solutions. This process helps in exploring the solution space and inheriting beneficial traits from the parents.

Crossover combines traits from two parent solutions P_a and P_b to generate new solutions:

$$M' = (M_a \cup M_b) \setminus (M_a \cap M_b) \quad (17)$$

$$\Delta' = \Delta_a \cup \Delta_b \quad (18)$$

where:

- M_a, M_b : Sets of pixel locations from the parent solutions P_a and P_b , respectively.
- M' : The set of pixel locations for the new solution.
- Δ_a, Δ_b : Sets of perturbation values corresponding to M_a and M_b .
- Δ' : The set of perturbation values for the new solution.

8.3. Mutation

The mutation operation introduces variations by modifying a subset of pixel locations. This process helps in exploring new regions of the solution space and maintaining diversity among solutions.

Mutation introduces variations to solutions by modifying a subset of pixel locations:

$$M'' = (M' \setminus A) \cup B, \quad \Delta'' = (\Delta' \setminus \Delta_A) \cup \Delta_B \quad (19)$$

where:

- M' : The set of pixel locations before mutation.
- A : A randomly selected subset of pixel locations to be replaced.
- B : A new subset of pixel locations to be introduced (obtained through random selection).
- M'' : The set of pixel locations after mutation.

- Δ' : The set of perturbation values before mutation.
- Δ_A : The perturbation values corresponding to A .
- Δ_B : The perturbation values corresponding to B .
- Δ'' : The set of perturbation values after mutation.

8.4. Evaluation

The evaluation process calculates the objective vector $\Phi(f_{adv}) = (\tilde{\mathcal{D}}(f_{adv}), \tilde{\mathcal{S}}(f_{adv}), \|\eta\|_2, \|\eta\|_0)^T$ for each solution to measure its quality. This step is crucial for selecting the best solutions.

8.5. Selection

Algorithm 2 Non-Dominating Sorting for Multiform Attack

```

1: Input: Combined population  $\mathcal{P}$ , objective vectors  $\Phi$ 
2: Output: Set of fronts  $\mathcal{S}_f$ 
3:  $\mathcal{S}_f \leftarrow \{\}$  // Initialize the set of fronts
4: for  $p \in \mathcal{P}$  do
5:    $S_p \leftarrow \{\}$  // Set of  $p$  dominated solutions
6:    $n_p \leftarrow 0$  // Domination counter of  $p$ 
7:   for  $q \in \mathcal{P}$  do
8:     if  $\text{DOMINATES}(p, q, \Phi)$  then //if  $p$  dominates  $q$ 
9:        $S_p \leftarrow S_p \cup \{q\}$ 
10:    else if  $\text{DOMINATES}(q, p, \Phi)$  then
11:       $n_p \leftarrow n_p + 1$ 
12:    end if
13:  end for
14:  if  $n_p == 0$  then
15:     $p_{\text{rank}} \leftarrow 1$  //  $p$  belongs to the first front
16:     $\mathcal{S}_{f1} \leftarrow \mathcal{S}_{f1} \cup \{p\}$ 
17:  end if
18: end for
19:  $i \leftarrow 1$  // Initialize front counter
20: while  $\mathcal{S}_{fi} \neq \emptyset$  do
21:    $Q \leftarrow \{\}$  // Store solutions of the next front
22:   for  $p \in \mathcal{S}_{fi}$  do
23:     for  $q \in S_p$  do
24:        $n_q \leftarrow n_q - 1$ 
25:       if  $n_q == 0$  then
26:          $q_{\text{rank}} \leftarrow i + 1$ 
27:          $Q \leftarrow Q \cup \{q\}$ 
28:       end if
29:     end for
30:   end for
31:    $i \leftarrow i + 1$ 
32:    $\mathcal{S}_{f(i+1)} \leftarrow Q$ 
33: end while
34: return  $\mathcal{S}_f$ 

```

Algorithm 3 Function to Determine if One Solution Dominates Another

```

1: function  $\text{DOMINATES}(x, y, \Phi)$ 
2:    $isBetter \leftarrow \text{False}$ 
3:   for  $i \leftarrow 1$  to  $\text{length}(\Phi)$  do
4:     if  $\Phi_i(x) > \Phi_i(y)$  then
5:        $isBetter \leftarrow \text{True}$ 
6:     else if  $\Phi_i(x) < \Phi_i(y)$  then
7:       return  $\text{False}$ 
8:     end if
9:   end for
10:  return  $isBetter$ 
11: end function

```

The selection process evaluates solutions and chooses the best individuals to pass their genetic material to the next

generation. Initially, non-dominated sorting is used to select the best solutions from the combined parent and offspring populations.

Selection uses non-dominated sorting to choose the best solutions for the next generation:

$$P' = \text{NonDominatedSort}(P \cup O) \quad (20)$$

where:

- P and O are the parent and offspring populations, respectively.
- P' : The new population of solutions after selection.
- NonDominatedSort : A sorting method that selects non-dominated solutions based on their objective vectors.

The selection process can be represented as:

$$\text{Selected Solutions} \eta = \arg \min_{s \in P'} \Phi(\mathcal{F}(x + \delta + \eta_s)) \quad (21)$$

where:

- P' is the set of non-dominated solutions selected by non-dominated sorting.

In this way, we first use non-dominated sorting to select a set of non-dominated solutions and then choose the solutions with the smallest objective function values from these non-dominated solutions.

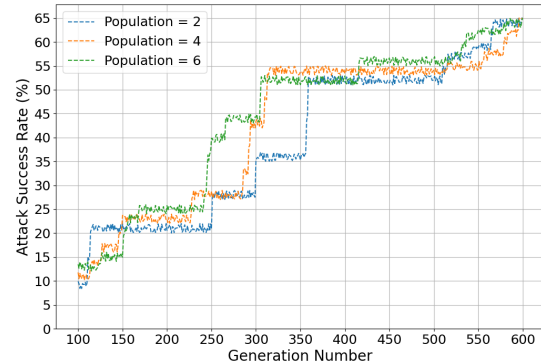


Figure 3. Attack success rate by generation number for different population sizes.

Fig.3 illustrates the relationship between the generation number, population size, and attack success rate during the evolutionary process. Since the primary goal of evolutionary search is to optimize the universal perturbation, we choose not to use the configuration with the highest attack success rate in practice to reduce time costs. Instead, we set the generation number to 150 and the population size to 2.

Table 6. Defense performance comparison between normal training and adversarial training using CMPS attack and our attack. Experiments were conducted on the RegDB dataset using AGW as the baseline. Rank-1 and mAP are reported.

Training Type	CMPS attack		Our attack	
	Rank-1 (%)	mAP (%)	Rank-1 (%)	mAP (%)
Normal Training	2.29	3.92	0.98	7.93
Adv. Training (CMPS attack)	35.61	30.43	28.12	24.56
Adv. Training (Our attack)	40.34	35.29	38.56	33.14

9. Adversarial Defense Analysis

9.1. Adversarial Training for Multimodal Robustness

To further validate the practical value of our proposed method, we integrate it into the adversarial training pipeline to investigate its efficacy in enhancing model robustness against multimodal attacks. Adversarial training, a widely used defense mechanism, strengthens model resilience by incorporating adversarial samples during the training process. This section explores whether adversarial training using adversarial samples generated by the proposed Multimodal Attack can significantly improve defense performance across heterogeneous modalities.

9.2. Experimental Setup

We conducted experiments on the RegDB dataset to evaluate the baseline model AGW. Adversarial samples were generated using our Gradient-Evolutionary Multimodal Optimization framework with $\epsilon = 8$. During adversarial training, each mini-batch consisted of 50% adversarial samples and 50% original samples. We compared our method with the state-of-the-art cross-modal attack method CMPS to test the model’s adversarial robustness. Performance was assessed using Rank-1 accuracy and mAP to evaluate the robustness of the trained models.

9.3. Results and Discussion

Tab. 6 presents the experimental results. From the table, we observe that models trained with normal training exhibit poor robustness to both CMPS and our proposed attack, achieving only 2.29% Rank-1 accuracy and 3.92% mAP under CMPS attack, and even lower results of 0.98% Rank-1 and 7.93% mAP under our attack. These results highlight the vulnerability of the baseline model to cross-modal adversarial attacks, especially under the stronger perturbations generated by our method.

When adversarial training is applied using CMPS attack samples, the model’s robustness improves significantly, achieving 35.61% Rank-1 accuracy and 30.43% mAP under CMPS attacks. However, under our attack, the performance is slightly lower, with 28.12% Rank-1 accuracy and 24.56% mAP. This demonstrates that adversar-

ial training with CMPS samples improves robustness to CMPS-specific attacks but provides limited transferability to stronger and more diverse attacks, such as those generated by our method.

Adversarial training using samples generated by our method further improves robustness across both attack types. The model achieves 40.34% Rank-1 accuracy and 35.29% mAP under CMPS attacks, demonstrating improved generalization against diverse adversarial strategies. Similarly, the model achieves 38.56% Rank-1 accuracy and 33.14% mAP under our attack, indicating the superiority of adversarial training with our method in preparing the model to handle stronger perturbations.

These results highlight the importance of adversarial training in enhancing cross-modal robustness. Furthermore, the Gradient-Evolutionary Multimodal Optimization framework not only generates highly effective adversarial samples but also provides a robust defense mechanism when used in adversarial training. This dual utility underscores the potential of our method to advance security and robustness in cross-modal ReID systems.

10. Feasibility Analysis of Evolutionary Search

Here, we simplify the problem and appropriately reformulate it to facilitate analysis.

10.1. Objective of Sparse Perturbations

The goal of introducing sparse perturbations is to minimize the number of changes made to the input while maximizing the error induced across various models. The fitness function, which evaluates the effectiveness of a perturbation across multiple models, is central to this process.

10.2. Fitness Function

Let Φ represent a collection of models $\{M_1, M_2, \dots, M_k\}$, where each model M_i has an associated misclassification rate $r_i(x)$ for an input x . The fitness function $f(x)$ for a perturbation vector x is defined as:

$$f(x) = \sum_{i=1}^k w_i \cdot r_i(x) - \lambda \cdot \|x\|_0 \quad (22)$$

Here, $\|x\|_0$ denotes the sparsity of the perturbation vector (the number of non-zero elements), w_i is the weight associated with model M_i , reflecting its importance in the overall fitness, and λ is a regularization parameter that controls the significance of sparsity.

10.3. Generation of Sparse Perturbations

Evolutionary algorithms utilize selection, crossover, and mutation operations to generate new perturbations. The

mutation operation, designed to maintain sparsity while enhancing fitness, is defined as:

$$x' = x + \delta \quad (23)$$

where δ is a small change vector, typically non-zero in only a few components of x . δ is chosen to maximize $f(x + \delta)$.

10.4. Convergence Analysis

By appropriately choosing the parameters λ and w_i , evolutionary search can converge to high-quality sparse perturbations. Theoretically, consider the change in the fitness function over iterations. assuming:

$$\lim_{t \rightarrow \infty} f(x^{(t)}) = f^* \quad (24)$$

where $x^{(t)}$ is the perturbation vector after the t -th iteration, and f^* represents the optimal achievable fitness.

To ensure that each iteration does not decrease the fitness, we need:

$$f(x^{(t+1)}) \geq f(x^{(t)}) \quad (25)$$

This indicates that the fitness is non-decreasing over iterations, suggesting that the algorithm is at least locally optimal. By designing δ to ensure that $f(x + \delta) > f(x)$, this can be achieved. This can be specifically implemented through techniques such as non-dominated sorting to select the most effective perturbations. Please note that our goal is to fine-tune the universal perturbation using the search process; in fact, we do not need to find the perturbation with the optimal fitness value.

Through the analysis provided, we theoretically demonstrate that evolutionary search can generate effective sparse perturbations that are effective across multiple models.

11. Feasibility Analysis of Enhancing Attack Transferability

Here, we simplify the problem and appropriately reformulate it to facilitate analysis.

11.1. Objective Definition

The goal is to find a fine-tuned perturbation δ_f that significantly increases the misclassification rate across multiple models while maintaining the sparsity of the perturbation. For this purpose, we introduce a fitness function:

$$f(\delta_f) = \sum_{i=1}^k w_i \cdot r_i(\delta_u + \delta_f) - \lambda \cdot \|\delta_f\|_0 \quad (26)$$

where $\|x\|_0$ denotes the sparsity of the perturbation vector (the number of non-zero elements), w_i is the weight associated with model M_i , reflecting its importance in the overall fitness, and λ is a regularization parameter that controls the significance of sparsity.

11.2. Complementarity Measure

To quantify the complementarity effect of the fine-tuned perturbation δ_f on the universal perturbation δ_u , we define the complementarity measure α_i as:

$$\alpha_i = \frac{r_i(\delta_u + \delta_f) - r_i(\delta_u)}{r_i(\delta_u)} \quad (27)$$

Here, α_i represents the increase in the misclassification rate of model M_i when the fine-tuned perturbation δ_f is added to the universal perturbation δ_u . Ideally, α_i should be significantly greater than 0, indicating a strong complementarity effect.

11.3. Optimization of the Fitness Function

The goal of the evolutionary search is to maximize the fitness function $f(\delta_f)$. By selecting appropriate mutation vectors δ and performing crossover operations, we ensure that each iteration finds a better fine-tuned perturbation δ_f . Specifically, we aim to find δ_f such that:

$$\sum_{i=1}^k w_i \cdot r_i(\delta_u + \delta_f) - \lambda \cdot \|\delta_f\|_0 \quad (28)$$

is maximized.

11.4. Theoretical Guarantee

Through selection, crossover, and mutation operations, evolutionary search can effectively optimize the perturbation vector δ_f to exhibit a high complementarity measure α_i across multiple models.

Specifically, for each iteration, we have:

$$\begin{aligned} \alpha_i^{(t+1)} &= \frac{r_i(\delta_u + \delta_f^{(t+1)}) - r_i(\delta_u)}{r_i(\delta_u)} \\ &\geq \alpha_i^{(t)} = \frac{r_i(\delta_u + \delta_f^{(t)}) - r_i(\delta_u)}{r_i(\delta_u)} \end{aligned} \quad (29)$$

This indicates that the complementarity measure of δ_f increases with each iteration, thereby significantly improving the misclassification rate across multiple models.

By appropriately choosing the parameters λ and w_i , evolutionary search can converge to high-quality sparse perturbations. Theoretically, considering the change in the fitness function over iterations, we assume:

$$\lim_{t \rightarrow \infty} f(x^{(t)}) = f^* \quad (30)$$

where $x^{(t)}$ is the perturbation vector after the t -th iteration, and f^* represents the optimal achievable fitness.

To ensure that each iteration does not decrease the fitness, we need:

$$f(x^{(t+1)}) \geq f(x^{(t)}) \quad (31)$$

This indicates that the fitness is non-decreasing over iterations, suggesting that the algorithm is at least locally optimal. By designing δ to ensure $f(x + \delta) > f(x)$, this can be achieved. Specifically, this can be implemented through non-dominated sorting to select the most effective perturbations.

By selecting appropriate mutation vectors δ and performing crossover operations, evolutionary search can effectively optimize the perturbation vector δ_f to exhibit a high complementarity measure α_i across multiple models, i.e.:

$$\alpha_i = \frac{r_i(\delta_u + \delta_f) - r_i(\delta_u)}{r_i(\delta_u)} \gg 0 \quad (32)$$

This indicates that the fine-tuned perturbation significantly increases the misclassification rate for each model, thereby improving the overall attack effectiveness.

Through the above mathematical analysis and theoretical derivation, we demonstrate that evolutionary search can find fine-tuned perturbations δ_f with a high complementarity measure α_i , effectively refining the universal perturbation δ_u , thus achieving better transferability and attack effectiveness across multiple models. This method leverages the global search capability of evolutionary algorithms and the complementarity between different models, ensuring that the perturbations are effective across various models.

12. CnMix Processing Algorithm

Here we provide the detailed processing workflow of Cn-mix, which generates new images by randomly mixing the three channels of the images. For specific details, refer to Alg. 4.

13. Discussion

13.1. Ethical Considerations

In this study, we introduce a novel cross-modal adversarial attack method that enhances the transferability and concealment of adversarial attacks through a gradient-evolutionary multiform optimization framework. This research offers a new perspective on understanding and enhancing the security of cross-modal systems, but it also raises a series of ethical and safety concerns about the potential negative impacts of adversarial attack techniques. Adversarial attack technology can be maliciously exploited, posing a serious threat to public safety.

However, we recognize the positive value of adversarial attack research. It reveals vulnerabilities in existing systems, prompting academia and industry to make in-depth

improvements to the robustness of machine learning models. The positive impact of this study lies in its potential to combine adversarial training with the attack methods presented to enhance system security and bring positive social impacts. Therefore, we emphasize the importance of conducting adversarial attack research within an ethical framework and encourage further development of defensive technologies to build a safer and more reliable technological environment.

13.2. Limitations and Future Work

Here, we need to acknowledge the limitations of the proposed method and identify potential areas for future research. Firstly, current attack techniques mainly focus on enhancing the stealthiness of perturbations for RGB images. In other types of images, such as infrared or thermal images in monochromatic modes, the invisibility of perturbations remains a challenge. This is because these modalities typically lack rich color information, making even subtle perturbations noticeable. Future research needs to explore how to effectively hide adversarial perturbations in these different image modalities to improve the applicability and stealthiness of attacks.

Secondly, as the number of modalities processed increases, the computational demand rises sharply, which can lead to significant resource consumption in practical applications. The high demand for computational resources in current methods limits their feasibility in resource-constrained environments. Therefore, future work could explore more efficient algorithms by incorporating state-of-the-art evolutionary computation methods to reduce the computational burden when handling large-scale or multi-modal data.

Algorithm 4 Random Channel Mixing of Grayscale and Sketch Images

```

1: Input: img (Input RGB image), G (Probability of converting to grayscale), Grgb (Probability of mixing grayscale and
   RGB), Srgb (Probability of mixing sketch and RGB)
2: Output: output_img (Transformed image)
3: function TOSKETCH(img)
4:   img_np  $\leftarrow$  Convert img to numpy array
5:   img_inv  $\leftarrow$  255 - img_np ▷ Invert image colors
6:   img_blur  $\leftarrow$  Gaussian blur of img_inv
7:   img_blend  $\leftarrow$   $\frac{img\_np}{255 - img\_blur} \times 256$  ▷ Blend original and blurred images
8:   return Convert img_blend to image
9: end function
10: function RANDOM_CHOOSE(r, g, b, gray_or_sketch)
11:   p  $\leftarrow$  [r, g, b, gray_or_sketch, gray_or_sketch]
12:   idx  $\leftarrow$  [0, 1, 2, 3, 4]
13:   Shuffle idx
14:   return Merge channels p[idx[0]], p[idx[1]], p[idx[2]] into RGB image
15: end function
16: function FUSE_RGB_GRAY_SKETCH(img, G, Grgb, Srgb)
17:   r, g, b  $\leftarrow$  Split img into RGB channels
18:   gray  $\leftarrow$  Convert img to grayscale
19:   p  $\leftarrow$  Random value between 0 and 1
20:   if p < G then
21:     return Merge gray, gray, gray into RGB image
22:   else if p < G + Grgb then
23:     output_img  $\leftarrow$  RANDOM_CHOOSE(r, g, b, gray)
24:     return output_img
25:   else if p < G + Grgb + Srgb then
26:     sketch  $\leftarrow$  TOSKETCH(gray)
27:     output_img  $\leftarrow$  RANDOM_CHOOSE(r, g, b, sketch)
28:     return output_img
29:   else
30:     return img
31:   end if
32: end function

```
